



Full length article

Assessing the chemical state of chemically deposited copper sulfide: A quantitative analysis of the X-ray photoelectron spectra of the amorphous-to-covellite transition phases



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ABSTRACT

Precise determination of the chemical composition and structure of copper sulfides is important to understand the different phase properties. To contribute in this matter, a set of peak-fitting parameters that allows for an accurate reproduction and subsequent quantitative analysis of the X-ray photoemission spectra of copper sulfides films is presented. The results show that the as-deposited brown-gold film, obtained by chemical solution deposition, is composed of Cu(I) and that it has an amorphous nature with a chemical composition of $\text{CuS}_{1.67 \pm 0.15}$, which is related to the composition of a pyrite-type structure. After annealing in a nitrogen-rich atmosphere, the film turned blue-green and shows a mixed Cu(I)/Cu(II) state with photoemission signals characteristic of covellite where the chemical composition was determined to be $\text{CuS}_{1.00 \pm 0.20}$. This chemical composition, in addition to the X-ray diffraction results, supports that the as-deposited amorphous copper sulfide film, when annealed, certainly turns into covellite. All the results suggest that the proposed quantitation method is suitable for quantitative analysis of copper sulfides, even when presenting an oxide overlayer.

1. Introduction

Determination of the copper (and sulfur) oxidation state in copper sulfides is still a debated issue. Uncertainty in this matter goes hand in hand with the uncertainty in the real chemical composition and structure of copper sulfide materials, for which at least nine defined structural phases are known to exist [1]: chalcocite (Cu_2S), djurleite ($\text{Cu}_{1.94}\text{S}$), digenite ($\text{Cu}_{1.80}\text{S}$), anilite ($\text{Cu}_{1.75}\text{S}$), geerite ($\text{Cu}_{1.60}\text{S}$), spionkopite ($\text{Cu}_{1.39}\text{S}$), yarrowite ($\text{Cu}_{1.12}\text{S}$), covellite (CuS), and a pyrite-type copper sulfide cubic phase (CuS_2); conveniently, these phases are better referred in literature as Cu_xS . This diversity of phases makes copper sulfide an interesting material not only for technological areas, but also for fundamental research, this because of the need to understand the formation of the various stoichiometries of these materials. The reason for the latter is that the various stoichiometries of copper sulfide, each of which is assigned to a specific structural phase, present clearly distinct electrical and optical properties [2–7], which definitively define

the ultimate application.

Although copper sulfide failed to form a stable solar cell structure when coupled with cadmium sulfide (CdS), which was the driving force for the research on copper sulfides during its first years of study (with the discovery of the photovoltaic effect of this structure [8]), nowadays there is a renewed interest due to the current technological applications of these materials. For our particular case, for example, we are interested in the use of chemical-solution-deposited copper sulfide thin films as a precursor for the copper-zinc sulfide (CZS) ternary and copper-zinc-tin sulfide (CZTS) quaternary semiconducting compounds for applications in photovoltaics, as well as a precursor of copper iodide (CuI) films [9] for application as a hole-transporting layer (HTL), which is also important in photovoltaics research. However, given that the electrical and optical properties largely depend on the structure and composition of the copper sulfides [2–7], the synthesis of these materials and the fabrication of devices (photovoltaic devices, in our case) should be done under well-defined and well-controlled conditions. In

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the same way, determining the real chemical composition of each copper sulfide phase is quite important to understand not only the previous properties, but also the phase stability and the possible interactions with the ambient gases and with other materials in the conformation of a particular device, which is essential for the long-performance of the same.

Given the above, it is clear that there is a lot of work to do regarding copper sulfides, especially in terms of the quantitative determination of the chemical state and composition of these materials. Therefore, this report deals with a proper chemical assessment using X-ray photoelectron spectroscopy (XPS) of copper sulfides obtained by chemical solution deposition. The present contribution proposes a set of peak-fitting parameters that enables a close reproduction of the photoemission spectra of copper sulfides and allows for the determination of peak areas that are employed in chemical composition calculations. A main feature of this quantitative approach is that the peak-fitting procedure is conducted employing the whole Cu $2p_{3/2}$ and Cu $2p_{1/2}$ spectral range, which is of utter importance in the assessment of peak intensities [10,11]. To the knowledge of the authors, reports analyzing the complete Cu 2p spectra are scarce. Another important feature is that the background trend on both sides of the photoemission spectra is accurately reproduced by means of the state-of-the-art practical background modeling approach and the use of the Shirley-Vegh-Salvi-Castle (SVSC) background [12–14]. Finally, the chemical composition is determined using an approach that takes into consideration that the photoemission signal is attenuated due to scattering, and the results provide an insight into the amorphous-to-covellite transition phases of copper sulfides. The intention of the present report is that the data presented can aid in theoretical studies regarding the intricacies of the electronic structure of copper sulfides as well as to provide a set of peak-fitting parameters applicable in quantitative studies of other copper sulfide phases.

2. Experimental and analysis details

2.1. Preliminary considerations

In the present research we analyze a brown-gold copper sulfide amorphous phase obtained by chemical solution deposition, which can be better associated to a pyrite-type structure (as discussed in detail later on); this copper sulfide quickly transforms to a blue-green copper sulfide crystalline phase after annealing at temperatures above 100 °C, being the resulted product associated to a covellite structure. Although the study of this phase transition [1,6] seems to be very particular, it should be noted that the present chemical assessment was made with the sake to report chemical parameters that are consistent and fully applicable to resolve XPS experimental data of most of the possible copper sulfide phases.

Importantly, to avoid inconsistencies and confusion in the interpretation of the XPS data, which seems to be common in literature, we attended the following experimental and measuring considerations:

1. The Cu(II) ions are reduced to Cu(I) in aqueous reaction solutions prior to precipitation of the copper sulfide material. Therefore, the precipitate or the film formed by means of this process should contain Cu(I). This matter was discussed by Luther et al. [15]. The amount of Cu(I), either as a sole chemical state or a mix with Cu(II), should be determined.
2. Obtaining a particular copper sulfide phase by chemical solution deposition depends on the deposition conditions, such as the concentration of reagents, kind of reagents, pH, reaction temperature, and deposition time. Some copper sulfide phases are differentiated by their color [5], which is a first indication that different structures have been obtained [1]. It is necessary, therefore, to define one particular formulation and to maintain deposition conditions highly controlled.
3. Crystallization of the chemical-solution-deposited amorphous

copper sulfide occurs when the film is heated. The change in color of the films from brown-gold to blue-green clearly evidences the phase transition addressed in this paper. This change can easily occur even when drying the fresh deposited films with a heat gun. Therefore, the films must be dried at room temperature in air or with a heat gun operated in cold mode. Additionally, careful detail should be taken with the variation of temperature in laboratory and transportation to the characterization facilities.

4. Transmission electron microscopy (TEM) is not appropriate to analyze the crystalline degree of copper sulfide materials, since the electron beam favors *in situ* changes in chemistry and structure during the measurement, as we observed in preliminary experiments. This is not particularly useful to determine possible low-range crystalline domains in samples that are amorphous to the X-ray diffraction technique, unless immediate images are taken. When using TEM, we clearly observed the crystallization of the amorphous copper sulfide in a short time during the measurements.
5. Oxidation of the copper sulfide surface occurs readily during the first 48 h after exposure to ambient. This leads to the formation of an overlayer of copper oxide, hydroxide, and/or sulfate species. This oxidized layer is likely composed of Cu(II), and the amount of formed Cu(II) depends on the phase structure and composition of the copper sulfide material. This matter was discussed by Kundu et al. [16]. It is important to avoid oxidation and, if this occurs, the thickness of this oxidized layer should be accurately determined. Therefore, after deposition, the samples were placed in an inert atmosphere and transferred to the XPS analysis chamber after a standardized 24 h exposure to the ambient had elapsed. Although this exposure time certainly lead to the formation of an oxidized layer, the resulted thickness, as discussed later, is in the Angstroms range.
6. Cleaning of the copper sulfide surface with argon ion sputtering prior the XPS data acquisition alters the copper oxidation state [17]. Despite of this, there are still some reports applying this procedure [18,19]. If a clean surface or a depth profiling is required, cleaning must be done with argon ion clusters [20] or other type of gas cluster ion beam technology to prevent surface chemistry modification, and therefore, to prevent chemical misassignments.
7. When dealing with samples that are prone to X-ray induced reduction [21–23], it is important to keep close attention to the evolution of the spectrum during acquisition. It is necessary to assure that the shape of the spectrum does not change after each scan during the course of the measurement. For this case, we have monitored the spectra and for the minimal conditions necessary to achieve a good signal to noise ratio no changes were detected, which gave us a high assurance that the acquired data actually corresponds to the measured samples.

All these previous considerations arose during the preliminary experiments and contributed to the formulation of a precise experimental design, which is presented below. The reliability of the results presented in this work is the product of the careful synthesis of the materials and acquisition of data, as well as a careful analysis of the X-ray photoelectron spectra.

2.2. Preparation of as-deposited and annealed copper sulfide films

Copper sulfide films were obtained using the chemical solution deposition method. Stock solutions were prepared using anhydrous copper(II) sulfate, CuSO₄ (98% purity, Spectrum); triethanolamine (TEA), N(CH₂CH₂OH)₃ (99.8% purity, Fermont); ammonium hydroxide, NH₄OH (aqueous solution with 29% of NH₃, Fermont); and thiourea (TU), (NH₂)₂CS (99.4% purity, Fermont). The aqueous reaction solution employed to deposit the copper sulfide films was composed of 0.01 M CuSO₄, 0.16 M TEA, 0.24 M NH₄OH, 0.02 M NaOH, and 0.03 M TU. The resultant pH of this reaction solution is ~10, as tested

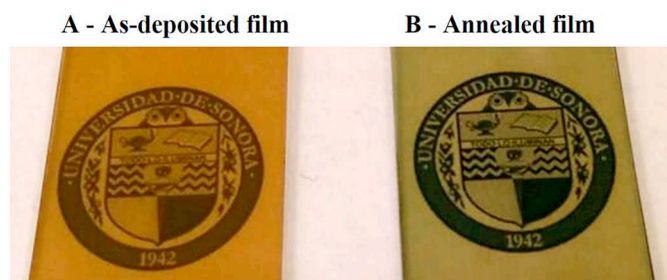


Fig. 1. Photograph of the copper sulfide films where A corresponds to an as-deposited film while B represents an annealed film. There is a clear difference in the color of both films where the as-deposited has a brown-gold color and the annealed film possess a blue-green color. The logo of the Universidad de Sonora is presented in the background.

with a MColorpHast™ pH-indicator strip. The total volume of the reaction solution employed for the deposition process was 95.5 mL. Details on the preparation of this reaction solution are also reported in a parallel work focused on processing and transformation of copper sulfide thin films [9].

The films were deposited on VELAB™ microscope glass slides substrates (2.5 cm × 7.5 cm × 0.1 cm) that were thoroughly cleaned with Alconox® detergent and distilled water and rinsed later with abundant distilled water. The deposition process consisted in vertically immersing the clean glass substrates in the reaction solution described before for 40 min at a reaction temperature of 50 °C; these reaction conditions were selected to obtain a copper sulfide thin film with a thickness of about 105 nm. At the end of this deposition time, the substrates coated with the deposited copper sulfide film were withdrawn from the reaction solution, rinsed with distilled water, later with isopropanol, and then dried with a heat gun in cold mode. The color of these films resulted brown-gold, as appreciable in Fig. 1-A.

Another set of as-deposited films were annealed inside a glovebox system from Inert® that was flooded with ultra-high purity molecular nitrogen, an oxygen level of < 0.1 ppm, and a moisture level of 30 ppm. The glovebox was maintained at atmospheric pressure. The annealing was done with an IKA® hot plate at 140 °C for 10 min. An example of the resultant films, that show a blue-green color, is presented in Fig. 1-B.

2.3. Thickness measurement and structural characterization

The thickness of the as-deposited copper sulfide film was measured with an Alpha-Step 100 (Tencor, CS) stylus profilometer. The crystallinity of the as-deposited and annealed films was probed by X-ray diffraction in grazing incidence geometry using a Rigaku DMAX-2200 diffractometer that employs Cu Kα ($\lambda = 1.54 \text{ \AA}$) radiation and the incidence angle was set to 0.5°.

2.4. Acquisition of X-ray photoelectron spectra

The experimental X-ray photoelectron spectra measurements were carried out in a Thermo Scientific™ ESCALAB™ 250Xi XPS system equipped with a monochromatic Al Kα₁ ($h\nu = 1486.7 \text{ eV}$) XR6 X-ray source and a 180° hemispherical energy analyzer with an array of six channel electron multipliers. The background pressure during data acquisition was $\sim 10^{-10}$ Torr. The standard lens operation mode was employed, and the monochromator was set to a 650 μm spot size. The system calibration was done following the International Standard ISO 15472 [24], where the work functions of the system were set to give a metallic gold Au 4f_{7/2} binding energy peak of 83.96 eV. Additionally, the binding energy scale is adjusted by setting the metallic Cu 2p_{3/2} and Ag 3d_{5/2} peaks to binding energies of 932.62 and 368.21 eV, respectively. The pass energy of operation was 10 eV that corresponds to an overall Ag 3d_{5/2} full width half maximum (FWHM) of 0.48 eV. Charge

compensation was done employing an electron flood source and a small energy correction of around 0.8 eV was needed to set the adventitious C 1s peak to 284.80 eV. The take-off angle was set to $\theta = 90^\circ$ with respect to the sample surface. The X-ray exposure was limited during the measurements by setting a minimal number of scans and total acquisition times not exceeding 1 h per sample. This condition was attended to assure that no chemical modification of the surface sample occurs (according to Point 7 of the list presented in the subsection of Preliminary considerations).

2.5. Quantitative analysis

The approach employed to quantify the X-ray photoelectron spectra of copper sulfide films is based on assessing the intensity of the photoemission peaks via accurate peak-fitting and using physical quantities to calculate the chemical composition of the films. This approach has been employed in several recent reports involving the photoemission spectra of metal oxides [10,11,25–28] and tin sulfide [29], copper iodide [9], and even glass surfaces [30].

The peak-fitting analysis was done using the AAnalyzer® software [31]. An important aspect of the analysis is that the active approach is employed for background modeling [14]. This state-of-the-art approach allows for the simultaneous consideration of several background contributions; in this case, a linear, a Shirley-type (SVSC) [14], and a Tougaard-type (slope-background [32]) background are employed and their individual contributions to the experimental spectra are determined during the optimization procedure.

The photoemission peaks were modeled using symmetric Voigt profiles. The theoretical branching ratio and Lorentzian widths were employed as fixed parameters [33,34]. The rest of the peak parameters, such as spin-orbit splitting, peak center, and area were evaluated as free variables and determined in the peak-fitting procedure. The determination of the Gaussian widths was done by constraining certain peaks to have the same value, under the assumption that peaks having a similar nature will have the same width (for example, mainline peaks have the same Gaussian width, loss peaks possess the same width, and satellites share the width value, where the widths for each set of peaks are evaluated as free variables). The uncertainties on the free parameters were evaluated using a rigorous approach that allows the determination of the peak parameters with a 95% confidence [25,35]. The aforementioned methodology for peak-fitting was employed because it has been also successfully applied in the analysis of the Cu 2p spectra of copper oxides [36].

Chemical composition was calculated employing the intensity of the photoemission peaks (peak areas) and a set of physical quantities to assess the atomic density of each present element in the films. The calculation considers that the films are homogenous and that the intensity of the *nl* core-level from each element *i* is proportional to its atomic density according to the following expression [10,11,37–39]:

$$I_{i,nl} \sim c \rho_i \frac{d\sigma_{i,nl}}{d\Omega} A_i \lambda_i \sin \theta_T \quad (1)$$

where $I_{i,nl}$ is the experimental peak area of the *nl* core-level of element *i*; *c* is a constant that accounts for the geometrical factors of the instrument [39]; ρ_i is the atomic density of each element; $d\sigma_{i,nl}/d\Omega$ is the differential photoelectric cross-section [40]; A_i is the efficiency of the spectrometer at the corresponding kinetic energy; λ_i is the effective attenuation length; and θ_T is the photoelectron take-off angle. The physical parameters required to evaluate the chemical composition were obtained from theoretical and tabulated values [41–43].

3. Results and discussion

3.1. The photoemission spectra of the as-deposited film

The Cu 2p photoemission spectrum of an as-deposited copper sulfide

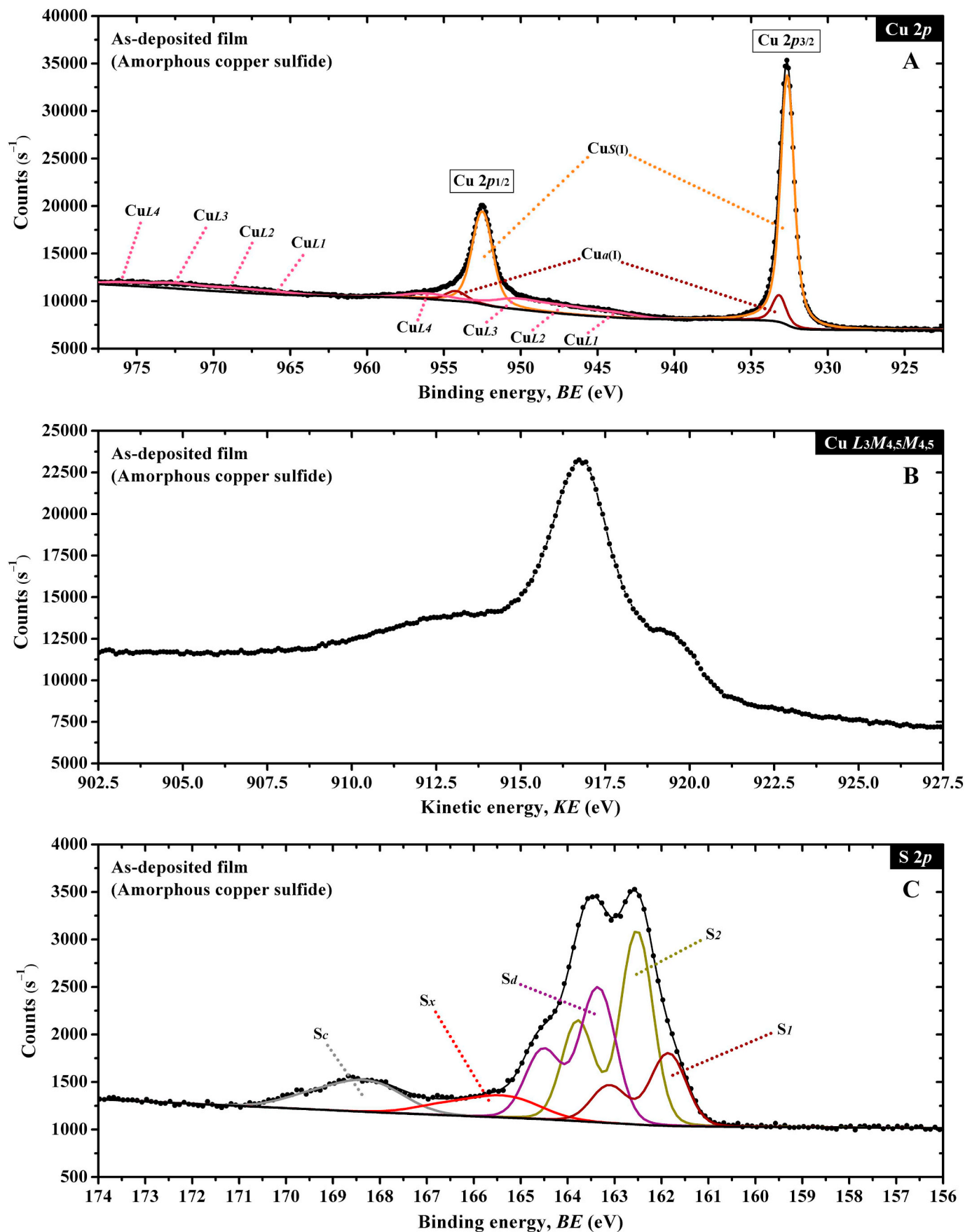


Fig. 2. Photoemission spectra of the as-deposited copper sulfide film. Peak parameters employed to reproduce the experimental data can be found in Table 1.

film is presented in Fig. 2-A. The peak-fitting analysis shows that it is possible to achieve a close experimental data reproduction over the whole fitting range that includes the $2p_{3/2}$ and $2p_{1/2}$ photoemission lines; this is an important result because most of the quantitative studies in the literature deal with the main and satellite peaks of the Cu $2p_{3/2}$ photoemission line only [44–46]. Indeed, there are many successful reports of quantitative analysis involving the sole analysis of the Cu $2p_{3/2}$ photoemission line, where the well-known satellite peaks are employed to discern between different copper oxidation states [21,44,45]; the method can in some circumstances be favorable when the signal intensity of the Cu $2p_{1/2}$ has a low signal intensity (when copper is not a major constituent of the sample), when there exists a strong background signal at the higher binding energy side of the spectra, when there are strong overlapping O KLL Auger lines, and when low acquisition times are required to prevent the X-ray induced reduction of Cu(II) [21–23]. However, the method is still prone to errors associated to the proper determination of the background signal, the pre-requisite of having reliable reference samples to assess the main peak/satellite area ratio, and the direct comparison between Cu $2p_{3/2}$ peak intensity with other photoemission peaks is not always straightforward.

Therefore, achieving an accurate peak-fitting over the whole Cu $2p$ fitting range is an aspect that deserves attention because this practice allows for a proper determination of the peak intensities [11,25]; especially when the background signal is complex and there exists a difference between the calculated background and the experimental signal in the region between the Cu $2p_{3/2}$ and Cu $2p_{1/2}$ core lines. This, however, turns out to be a major setback because analyzing the whole Cu $2p$ spectrum with an accurate determination of the background signal is complicated, but this task can be achieved using the active background approach. As mentioned before, another issue is the possible X-ray induced reduction of Cu(II), and consequently, the whole Cu $2p$ spectrum was monitored during acquisition to quickly detect a possible change in the shape of the spectrum with no apparent reduction being detected. It is important to mention that, the purpose of solving the setbacks of analyzing the entire Cu $2p$ spectrum is to assess peak intensities of copper and anion core-level peaks that allow an accurate determination of the chemical composition of the samples.

The peak parameters that were employed to reproduce the whole experimental spectrum can be found in Table 1. It is important to note that these peak-fitting parameters were determined through the use of state-of-the-art practical background modeling, that is, the active background approach [14] and by using Voigt profiles. The goodness of the peak-fitting results can be verified by the uniqueness plots [47] found in Appendix A.

From visual inspection, it is clear that the spectrum of the as-

deposited sample has the characteristics of a Cu(I) species due to the presence of an apparent single sharp peak and absence of the well-known satellite peaks of Cu(II) [16,44,45,48–53]. The results indicate that this copper sulfide film has the Cu(I) oxidation state, as it has been reported for many copper sulfides [50–53]. In this report, the main photoemission line is located at a binding energy of 932.60 ± 0.01 eV (peak $Cu_{S(I)}$) which again well agrees with other reports of copper sulfides with a Cu(I) state [45,46,52].

Additionally, a small intensity peak is found, whose binding energy position corresponds to 933.14 ± 0.02 eV (peak $Cu_{a(I)}$). This signal interestingly has a spin-orbit splitting value that is different from the main peak, which may lead to think that its presence is related to a satellite arising from Cu(I) with a mixed ground state configuration, that apart from the predominant $3d^{10}$, has small amounts of $3d^{10}4s\bar{s}$ (where $\bar{s}$ represents a hole in a sulfide ligand) and $3d^94s$ character, as suggested by Goh et al. [53], and others [1,54,55]. However, from its sharpness and energy separation from the main peak, the latter assumption seems unlikely. Therefore, another interpretation is that peak $Cu_{a(I)}$ may be related to an asymmetric component of the main Cu(I) photoemission line, where the apparent asymmetry may be related to the excitation of electrons near the Fermi level [56,57]. Additionally, the possibility of this signal in turn being actually related to a copper defect state or to impurities and surface contamination [58] cannot be discarded.

The remaining peaks of the Cu $2p$ photoemission spectrum (Cu_{L1} , Cu_{L2} , Cu_{L3} , and Cu_{L4}) can be associated to energy-loss features. Despite that the spin-orbit splitting values do not match the one of the main photoemission line, the intensity of the SVSC parameter of peaks Cu_{L1} , Cu_{L2} , Cu_{L3} , and Cu_{L4} , which is one order of magnitude greater than that of peak $Cu_{S(I)}$, suggests that these peaks are energy-loss features and are actually part of the background and not needed in chemical composition calculations [25].

The Cu $L_{3M_{4,5}M_{4,5}}$ Auger spectrum is presented in Fig. 2-B; from a qualitative standpoint, the spectrum bears close resemblance to that of copper sulfides, especially to that of Cu_2S [16,45,46,53]. The Auger peak with the largest intensity appears around 916.79 eV, and agrees to values reported for Cu_2S [46]. The chemical state of the as-deposited film was also assessed through the Auger parameter [59]; where the sum of the binding energy of the main Cu $2p$ photoemission line and the kinetic energy of the most intense Auger signal yields the true chemical state of the sample. The Auger parameter for the as-deposited film is found to be 1849.39 eV, also closely matching to copper sulfides [60]. The latter results do not necessarily imply that the as-deposited film is, in fact, Cu_2S ; but rather that the copper atoms in the as-deposited film have a similar local chemical environment with comparable electronic polarizability to that of copper atoms in Cu_2S .

Table 1

Peak parameters employed in the peak-fitting analysis of the photoemission spectra of the as-deposited copper sulfide film. The calculated uncertainties have a 95% confidence.

Core level	Peak label	$2p_{3/2}$ BE (eV)	Spin-orbit splitting (eV)	Peak width FWHM		SVSC (eV^{-1})	Normalized peak areas
				Gaussian (eV)	Lorentzian (eV)		
Cu $2p$	$Cu_{S(I)}$	932.60 ± 0.01	19.86 ± 0.01	0.62 ± 0.05	0.6	0.027	100.0 ± 0.5
				1.18 ± 0.05^a			
	$Cu_{a(I)}$	933.14 ± 0.02	21.05 ± 0.05	0.62 ± 0.05	0.6	0.027	10.7 ± 0.5
				1.18 ± 0.05^a			
	Cu_{L1}	943.81 ± 0.14	22.03 ± 0.23	3.51 ± 0.15	–	0.136	4.9 ± 0.3
	Cu_{L2}	947.08 ± 0.18	21.84 ± 0.18	3.51 ± 0.15	–	0.136	9.3 ± 0.4
S $2p$	Cu_{L3}	950.34 ± 0.14	22.16 ± 0.14	3.51 ± 0.15	–	0.136	10.1 ± 0.6
	Cu_{L4}	956.00 ± 0.11	19.97 ± 0.22	3.51 ± 0.15	–	0.136	6.0 ± 0.6
	S_1	161.84 ± 0.06	1.26 ± 0.08	0.77 ± 0.06	0.07	0.007	41.7 ± 8.0
	S_2	162.51 ± 0.01	1.26 ± 0.05	0.77 ± 0.06	0.07	0.007	100.0 ± 15.0
	S_d	163.34 ± 0.02	1.16 ± 0.08	0.83 ± 0.05	0.07	0.007	76.9 ± 6.0
	S_x	165.25 ± 0.25	1.30 ± 0.36	1.58 ± 0.12	0.07	0.007	21.3 ± 3.0
	S_c	168.15 ± 0.06	1.20 ± 0.09	1.58 ± 0.12	0.07	0.007	29.4 ± 2.0

^a Value corresponding to the $2p_{1/2}$ photoemission line.

The S 2p photoemission spectrum can be found in Fig. 2-C; again the experimental data resembles to that of copper sulfides, but in this case the spectrum of CuS₂ [57] bears closer similarities. The peak-fitting results show that three strong photoemission lines are present; the first signal is energy located at a binding energy of 161.84 ± 0.06 eV (peak S₁) which may be related to a sulfide component S²⁻ [1,16,50,61], the peak at a binding energy of 162.51 ± 0.01 eV (peak S₂) can be associated to a disulfide (S₂)⁻ signal [57,62], and the component located at 163.34 ± 0.02 eV (peak S_d) may be due to a defect non-stoichiometric sulfide component (S_x)²⁻ [1,16,61,63,64].

Additionally, there exists the presence of a peak located at 165.25 ± 0.25 eV, labeled S_x, which may be attributed to polysulfides (S_x)ⁿ⁻ [61,63], elemental sulfur [65,66], or even a by-product of the chemical solution deposition reaction related to sodium sulfite [67] (the presence of sodium is confirmed from the survey scans presented in Appendix B), and other oxoanions of sulfur [68]. However, the assignment for this peak still requires a more detailed understanding of the reaction mechanism of the copper sulfide films. The peak located at 168.15 ± 0.06 eV, labeled S_c, is mainly related to sulfates coming from the solution and even the oxidation of sulfur at the surface of the film [16,69–71]. It is striking, that these sulfur components (peak S_x and S_c) do not change in energy position (Table 3) even after the annealing treatment. Thus, these components are not related to the copper sulfide films but are part of surface contamination or by-products of the deposition reaction.

The chemical composition of the as-deposited copper sulfide film was calculated employing the intensity of the peaks Cu_{S(I)}, Cu_{a(I)}, S₁, S₂, and S_d. The assessed chemical composition yielded CuS_{1.67 ± 0.15}, meaning that the as-deposited copper sulfide film may have a pyrite-type chemical structure. The atomic percent for each photoemission signal employed to calculate the chemical composition can be found in Table 2. To assess the possible crystalline structure, Fig. 3 shows the X-ray diffraction measurements of the copper sulfide films, where the patterns show that the as-deposited film is predominantly amorphous (a result that is consistent to previous reports of chemically deposited copper sulfides [6]). This may indicate that the local environment of this amorphous copper sulfide is comparable to that of an amorphous pyrite-type CuS₂.

There are many complications related to the understanding of the stoichiometry and structure of amorphous copper sulfides, mainly due to its metastable nature that makes the proper assessment of the film elusive. However, according to the resolved photoemission lines and the assessed chemical composition it is possible that the amorphous as-deposited film has a chemical composition that can be ascribed to Cu⁺ (S₂)⁻ [50,62] with traces of sulfide and polysulfide components related to its phase transformation to the more stable covellite phase [1,6,16,61].

3.2. The photoemission spectra of the annealed film

After annealing, the copper sulfide film suffers apparent changes that can be related to changes in the chemical state of the film. As noted

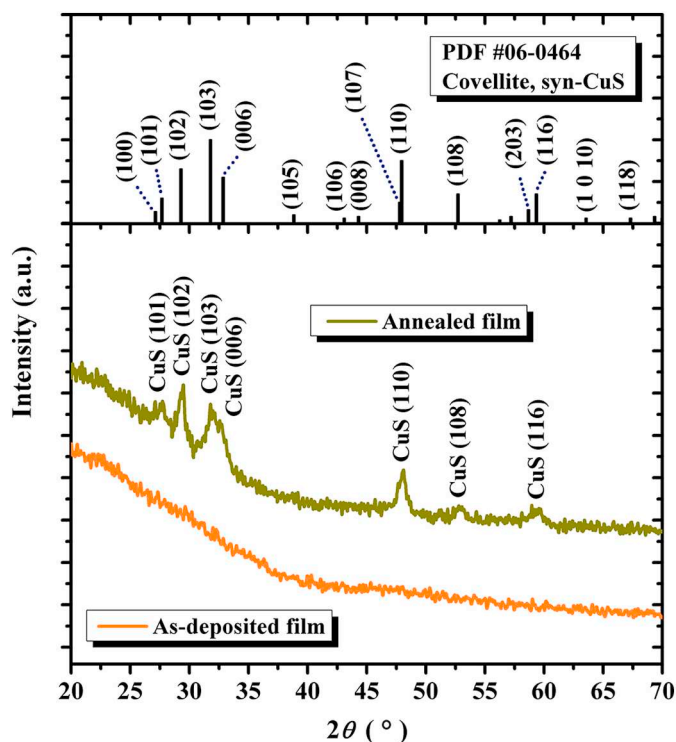


Fig. 3. X-ray diffraction of the copper sulfide films. The as-deposited copper sulfide film is found to be amorphous, while the annealed film shows diffraction peaks characteristic of covellite.

in previous reports [1,6], copper sulfide shows clear changes in color during an ageing or annealing treatment. In this report, the as-deposited film has a brown-gold appearance, and after the annealing treatment, the color of the film turns blue-green (as shown in Fig. 1). As it can be seen in Fig. 3, the change in color can be related (directly or indirectly) to the crystallization of the amorphous as-deposited film into a covellite copper sulfide film. Knowing that the annealed film has a covellite crystalline structure and by the apparent change in color, it is expected that the photoemission spectra of the annealed film also show clear differences when compared to the recorded spectra of the as-deposited film.

The Cu 2p photoemission spectrum of the annealed film is presented in Fig. 4-A; the differences between the as-deposited and the annealed film are striking to the eye because the appearance of a conspicuous large signal located at a binding energy of 933.26 ± 0.09 eV (peak Cu_{S(II)}). This signal can be related to Cu(II), mainly due to its energy position and because it is accompanied by characteristic satellites that are located in the energy range of 944–941 eV, these being the signature of Cu(II) [16,48–50]. It is noteworthy that in spite of Cu(I) being considered to be the main oxidation state of covellite [16,52,53,72,73] (whose presence is confirmed here with the appearance of the large intensity peak Cu_{S(I)}), in this report the experimental data strongly suggest that the annealed copper sulfide film has a mixed Cu(I)/Cu(II) state, a result that has been recently suggested for chemically deposited copper sulfide films [61]. In addition, it is interesting to note that the peak related to Cu(I), denoted here as Cu_{S(I)}, has shifted to 931.93 ± 0.01 eV, agreeing with values reported for covellite [53]. This may be related to the change in the local environment of Cu(I) atoms upon crystallization of the copper sulfide film, that is, a change from a pyrite-type to a covellite-type local environment.

The Cu 2p spectrum also shows peaks that may be related to the appearance of copper oxide compounds, in accordance to a report dealing with the effect of ambient atmosphere on copper sulfide films [16]. In the present paper, the peaks appear at binding energies of 934.37 ± 0.17 (peak Cu_{O1}) and 935.76 ± 0.17 (peak Cu_{O2}), alongside

Table 2

Atomic percent for each photoemission signal of the as-deposited film. The values were determined using Eq. (1). According to data presented below, the $(S_1 + S_2 + S_d)/(Cu_{S(I)} + Cu_{a(I)})$ atomic ratio, meaning the S/Cu ratio, is 1.67 ± 0.15 .

Peak label	Atomic percent (%)
Cu _{S(I)}	33.82 ± 1.90
Cu _{a(I)}	3.61 ± 0.21
S ₁	11.94 ± 1.60
S ₂	28.63 ± 3.22
S _d	22.00 ± 1.67

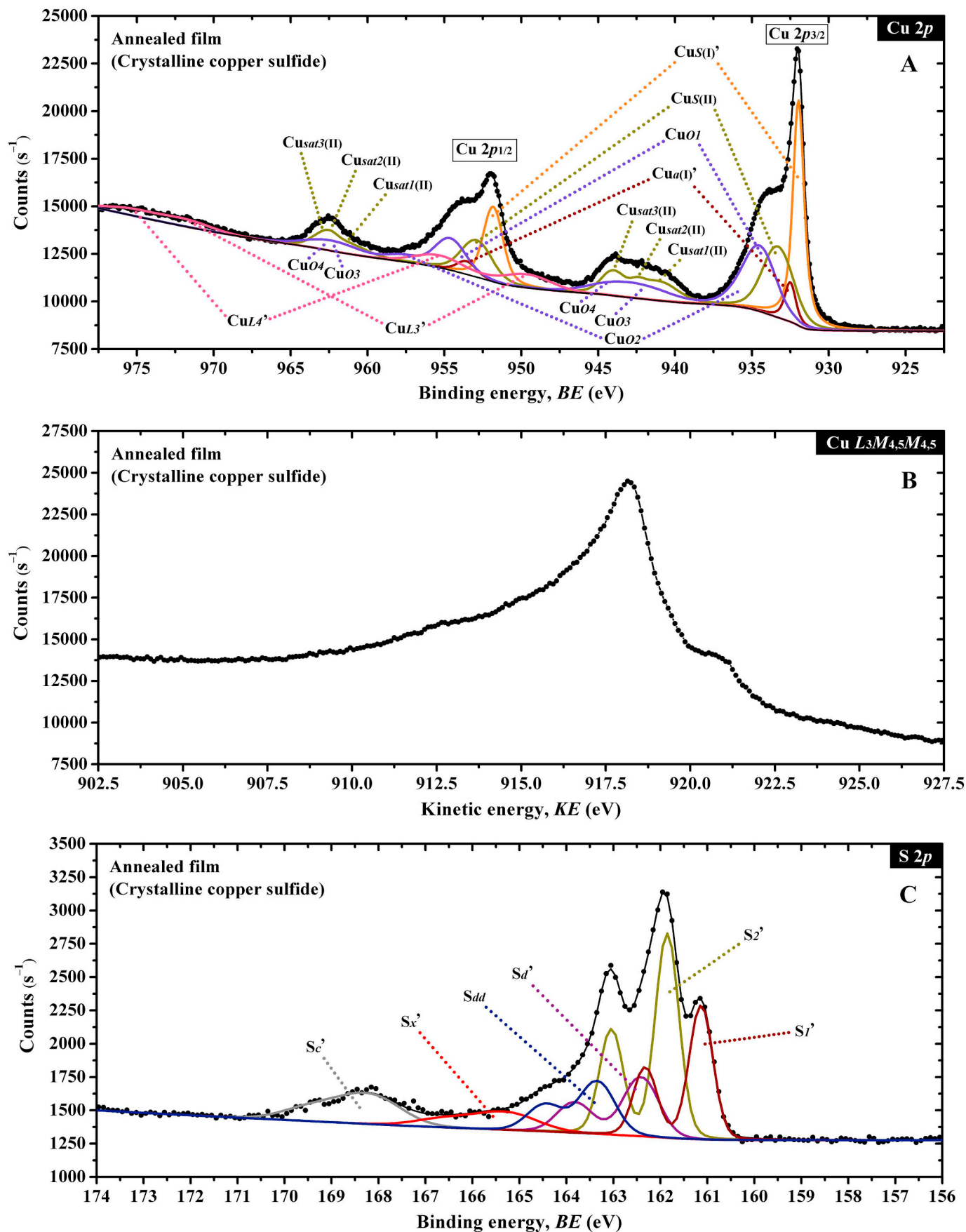


Fig. 4. Photoemission spectra of the annealed copper sulfide film. Peak parameters employed to reproduce the experimental data can be found in Table 3.

the presence of peaks in the energy range of 944–941 eV (peaks $\text{Cu}_{\text{O}3}$ and $\text{Cu}_{\text{O}4}$), which overlap with the satellite peaks $\text{Cu}_{\text{sat}1(\text{II})}$, $\text{Cu}_{\text{sat}2(\text{II})}$, and $\text{Cu}_{\text{sat}3(\text{II})}$. The present results suggest that the surface of the annealed copper sulfide film, in contrast to the as-deposited film, is oxidized to form an overlayer of copper oxide and/or copper hydroxide during the 24 h of exposure to the ambient.

The $\text{Cu}_{\text{L}3}\text{M}_{4,5}\text{M}_{4,5}$ Auger spectrum of the annealed film is presented in Fig. 4-B; once more the spectrum has close resemblance to others reported elsewhere, and the results are consistent with covellite exposed to an ambient atmosphere [16]. The features of the spectrum are a large intensity Auger peak centered at a kinetic energy of 918.25 eV and a clear asymmetric component; the signals may account for the Cu (I) state of covellite and the Cu(II) state of copper oxides. The Auger parameter considering the signals related to covellite yields 1850.18 eV, which agrees to the value reported for covellite [16,45]. With this Auger parameter value, it is further confirmed that the local chemical environment of the as-deposited film has truly changed when subjected to the annealing treatment.

The S 2p photoemission spectrum of the annealed film presents strong differences compared to the spectrum of the as-deposited thin film. Fig. 4-C shows the spectrum and it is conspicuous that the experimental data has the apparent three peak feature of covellite [16,52]. The peak-fitting analysis shows that the experimental data can be accurately reproduced employing a set of six doublet peaks. At first glance, it is noteworthy that peaks S_x and S_c appear in the same binding energy position and with similar relative intensity in both types of analyzed samples. These suggest that these compounds are not related to the copper sulfide films but are related to surface components or by-products of the chemical reaction deposition, as mentioned earlier.

The rest of the S 2p spectrum is comprised of several sulfur species: the peak S_1' , located at a binding energy of 161.12 ± 0.02 eV, can be related to a sulfide S^{2-} state [52,61,64,72]; the peak S_2' , located at 161.84 ± 0.02 eV, may be related to a disulfide $(\text{S}_2)^{2-}$ state [52,61,64,72]; and the peaks S_d' and S_{dd} , located at 162.40 ± 0.45 and 163.33 ± 0.21 eV, respectively, can be ascribed to defect non-stoichiometric sulfides $(\text{S}_x)^{2-}$ [1,16,61,63,64]. It is noteworthy that these sulfide species, meaning peaks S_1' , S_2' , S_d' , and S_{dd} , appear at different binding energies when compared to the as-deposited film, which alongside the clear change in color and XRD results, further confirms that the chemical nature of the copper sulfide film changes after annealing.

As mentioned before, the annealed copper sulfide film has the presence of an oxide overlayer; hence, to determine the chemical state of this species it is necessary to analyze the O 1s spectra of both films. The results are presented in Fig. 5 and it can be observed that several peaks are present. First, peaks located at binding energies of 531.12 ± 0.43 eV (peak $\text{O}_{\text{O}2}$) and 531.65 ± 0.12 eV (peak $\text{O}_{\text{C,adv}}$) may be related mainly to adsorbed molecular oxygen (O_2) and carbon-related adventitious contamination, respectively. Here, it is interesting to note that the peak $\text{O}_{\text{O}2}$ (at 531.12 ± 0.43 eV) increases in intensity when the as-deposited copper sulfide film is annealed; this increase in intensity may be related to the uptake of molecular oxygen from the atmosphere with a preferred adsorption on the copper oxide surface [74] easily formed on the covellite film [16] obtained after annealing, meaning that this contamination is less adsorbed on the surface of the as-deposited copper sulfide film. A probable small contribution from carbonate and cyanate to the intensity of this peak $\text{O}_{\text{O}2}$, especially for the as-deposited film, is not discarded; this arises from the presence of nitrogen in the as-deposited film, as confirmed from the survey scans presented in Appendix B. Furthermore, there is the presence of a peak located at 532.50 ± 0.16 eV (peak O_c) that may be related to sulfite and oxoanions of sulfur. The peak located at 533.63 ± 0.20 eV (peak $\text{O}_{\text{H}2\text{O}}$) may be possibly due to adsorbed H_2O at the surface [74–76] or even $\text{C}=\text{O}$ species. Finally, the peak located at 535.46 ± 0.20 is related to the $\text{Na KL}_1\text{L}_{2,3}$ Auger signal.

Peaks related to the copper oxides are energy located at

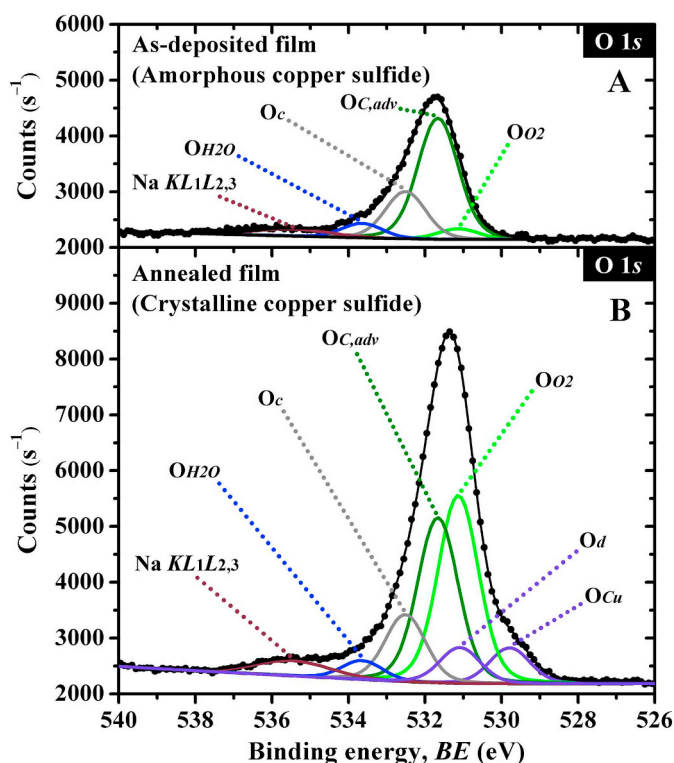


Fig. 5. O 1s photoemission spectra of the A as-deposited and B annealed copper sulfide film.

529.78 ± 0.10 (peak O_{Cu}) and 531.08 ± 0.50 eV (peak O_d), which coincide with oxygen bonded to copper in CuO , and defective low-coordinated oxide [77] or oxygen from copper hydroxide [45], respectively. The relatively large intensity of peak O_d might be related to a highly defective nanostructure resulting from the growth of substoichiometric native oxides as the surface is exposed to the atmosphere. The results show that only after the copper sulfide films are annealed, the formation of an oxide overlayer occurs, which implies that the chemical composition and structure of a given copper sulfide surface affects the oxygen uptake, as it is suggested by Kundu et al. [16]. Even more, this oxide overlayer promotes, in turn, the uptake of more oxygen from the atmosphere, as discussed above.

The determination of the chemical composition of the annealed copper sulfide needs an extra consideration due to the formation of an oxide overlayer. Therefore, the application of Eq. (1) is not straightforward and the attenuation of the photoemission signal due to scattering must be taken into account by following an approach similar to that employed for the quantification of a ZnO overlayer on a metallic zinc surface [25]. Here, Eq. (1) may be rewritten by considering that the oxide layer attenuates the copper sulfide photoemission signal in terms of a Beer-Lambert type factor:

$$I_{i,nl,s} \sim c \rho_{i,s} \frac{d\sigma_{i,nl}}{d\Omega} A_i \lambda_{i,s} \sin \theta_T \left[1 - \exp \left(-\frac{d_s}{\lambda_{i,s} \sin \theta_T} \right) \right] \prod_j^{\text{layers above } S} \exp \left(-\frac{d_j}{\lambda_{i,j} \sin \theta_T} \right) \quad (2)$$

In the latter set of equations, the subscript s denotes the physical parameters of a given photoelectron originated from a layer s ; thus, d_s is the thickness of layer s ; $\lambda_{i,s}$ denotes the effective attenuating length of a photoelectron coming from element i as it travels layer s ; d_j is the thickness of overlayer j ; and $\lambda_{i,j}$ is the effective attenuating length of a photoelectron coming from element i (element i located at layer s) as it travels layer j . The physical parameters employed to assess the chemical

composition of the annealed copper sulfide film (copper sulfide and copper oxide overlayer) can be found in Table 4.

Thus, the chemical assessment of the oxide overlayer shows that when the peaks ascribed to copper oxide in the Cu 2p and O 1s spectra (peaks Cu_{O1} , Cu_{O2} , Cu_{O3} , Cu_{O4} , O_{Cu} , and O_d) are employed, the calculation results in $\text{CuO}_{1.29 \pm 0.28}$, which indicates that copper(II) oxide grows on top of the covellite surface. The chemical composition, as well as the position and relative intensity of its satellites, confirms the presence of CuO. The thickness of this oxide (parameter $d_{j=ox}$ represents the thickness of the oxide overlayer on top of d_s in Eq. (2)), is determined to be around 0.5 nm, hence an oxide passivated layer on top of the covellite surface is formed. The latter is a result of assuming the formation of an overlayer and its description still requires additional analyses evaluating other possible nanostructures.

On the other hand, the present model takes into account a CuS_x substrate layer with infinite thickness (due to the probing depth of the XPS experiment $d_s = \infty$), and its chemical composition was calculated considering that the photoelectrons contributing to the area of peaks $\text{Cu}_{S(I)}$, $\text{Cu}_{a(I)}$, $\text{Cu}_{S(II)}$, $\text{Cu}_{sat1(II)}$, $\text{Cu}_{sat2(II)}$, $\text{Cu}_{sat3(II)}$, S_1' , S_2' , S_d' , and S_{dd} originate within this substrate layer. The chemical composition in terms of atomic percentage is presented in Table 5 and it can be observed that the photoemission signals associated to Cu(I) and Cu(II) have the same proportion ($24.93 \pm 3.87\%$ for Cu(I) and $25.07 \pm 7.17\%$ for Cu(II) in atomic percentage), which is an intriguing result because it does not comply with modern theoretical reports regarding the valences of copper and sulfur in covellite [78]; however, it is possible that the assessed annealed copper sulfide film has structural defects that tampers the relative proportion of Cu(I) and Cu(II) present in the film. The latter proposal may also explain the appearance of additional defect non-stoichiometric sulfides (S_x^{2-}), meaning that defects have a great impact in the resultant photoemission spectra of copper sulfides.

Furthermore, it is interesting to note that the total S/Cu ratio for the annealed copper sulfide film yields 1.00 ± 0.20 , a result that is in accordance to the expected stoichiometry of the CuS phase measured with XRD. However, it may be more relevant to analyze S/Cu ratios for each copper oxidation state. Thus, by considering the areas of the peaks $\text{Cu}_{S(I)}$, $\text{Cu}_{a(I)}$, S_1' , and S_2' , which are related to Cu(I) sulfide species, the chemical composition results in $\text{CuS}_{1.35 \pm 0.27}$; on the other hand, with the rest of the peak areas of $\text{Cu}_{S(II)}$, $\text{Cu}_{sat1(II)}$, $\text{Cu}_{sat2(II)}$, $\text{Cu}_{sat3(II)}$, S_d' , and S_{dd} , which are related to Cu(II) sulfide species, the calculation results in $\text{CuS}_{0.65 \pm 0.24}$. Even more, if the satellite peaks ($\text{Cu}_{sat1(II)}$, $\text{Cu}_{sat2(II)}$, and $\text{Cu}_{sat3(II)}$) are excluded, the total composition results in $\text{CuS}_{1.24 \pm 0.17}$, and the individual compositions yield $\text{CuS}_{1.35 \pm 0.22}$ (peaks $\text{Cu}_{S(I)}$, $\text{Cu}_{a(I)}$, S_1' , and S_2') for Cu(I) sulfide and $\text{CuS}_{1.06 \pm 0.28}$ (peaks $\text{Cu}_{S(II)}$, S_d' , and S_{dd}) for Cu(II) sulfide species. Due to the uncertainty associated to the determination of areas of closely lying peaks, the chemical composition results are not significantly different from each other. Therefore, solely from this approach it is not possible to discard the inclusion of the satellites, and the chemical analyses done show that including them in chemical composition calculations results in values that are in accordance to other characterization techniques, for example XRD and energy dispersive X-ray spectroscopy.

4. Conclusions

A chemical formulation that leads to copper sulfide thin films is

Appendix A. Uniqueness plots

Uniqueness plots are an excellent graphical tool to quickly assess the quality and reliability of the peak-fitting procedures and results obtained in the quantitative analysis of X-ray photoemission spectra [47]. Therefore, this appendix contains the uniqueness plots generated from the peak-fitting approach employed to analyze the Cu 2p and S 2p photoemission spectra of copper sulfides.

The uniqueness plot corresponding to the peak-fitting of the Cu 2p photoemission spectra of the as-deposited copper sulfide is presented in Fig. A1. Here the plot is constructed by plotting the reduced chi squared values of various fittings obtained upon the systematic variation of the Gaussian width of peaks $\text{Cu}_{S(I)}$ and $\text{Cu}_{a(I)}$. The plot shows that the reduced chi squared has a minimal value at the optimal Gaussian width of 0.62 eV and a noticeable curvature. Therefore, the optimal peak-fitting results presented in Fig. 2-A and Table 1 pass the uniqueness test making the results statistically meaningful.

presented, which, when as-deposited, are amorphous and brown-gold in color. The assessed chemical composition of these films yields $\text{CuS}_{1.67 \pm 0.15}$, thus indicating that these as-deposited copper sulfide films may have a pyrite-type chemical structure. In this sense, it is possible that the oxidation state of the amorphous as-deposited film is $\text{Cu}^+(\text{S}_2)^-$ with traces of sulfide and polysulfide components.

After annealing at 140 °C for 10 min, the copper sulfide film suffers clear changes that are related to changes in the chemical state, and thus, in its structure. First, its color changes from brown-gold to blue-green. The XRD results corroborated the structural change from amorphous, when as-deposited, to covellite, when annealed. The XPS experimental data strongly suggested that the annealed copper sulfide film has a mixed Cu(I)/Cu(II) state with a 1:1 ratio. The overall composition using the peak areas of the Cu 2p and the S 2p results in $\text{CuS}_{1.00 \pm 0.20}$, which validates the XRD measurements. However, the large relative proportion of defect non-stoichiometric sulfides indicates that the films have a large amorphous fraction.

XPS results also suggested the appearance of copper oxide compounds over the annealed sample. The thickness of this oxide overlayer was determined to be around 0.5 nm, and the chemical composition assessment resulted in $\text{CuO}_{1.29 \pm 0.28}$. Also, this overlayer was formed only on the annealed copper sulfide, which confirms that the chemical composition and structure of the surface affects the oxygen uptake.

A set of peak-fitting parameters that allows for an accurate reproduction and subsequent quantitative analysis of the X-ray photoemission spectra of copper sulfides films was presented as a valuable contribution for assessing chemical composition. The results presented along the body of this paper suggest that the proposed quantitation method is suitable for a quantitative analysis of copper sulfides phases.

As a final remark, it is important to mention that although we have presented carefully-determined peak-fitting parameters for an accurate reproduction of XPS experimental data for copper sulfides, it is still necessary to validate the assignment of the species proposed for each of them. This is a challenging task and should be addressed separately with appropriate theoretical chemical modeling.

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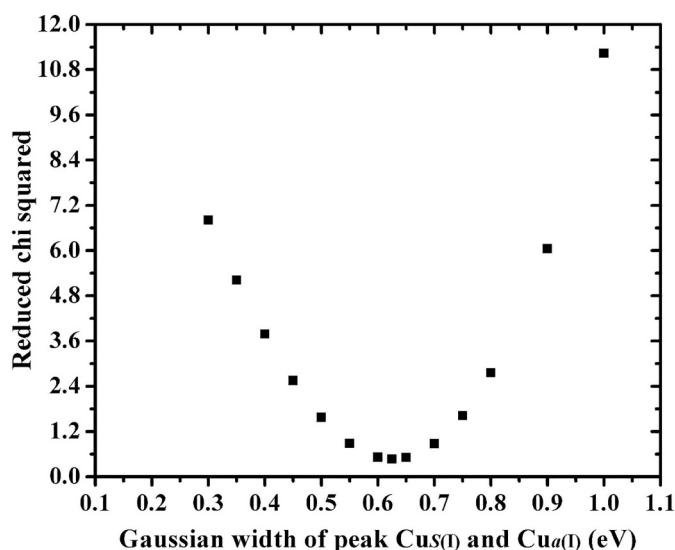


Fig. A1. Uniqueness plot obtained from peak-fitting the Cu 2p photoemission spectra of the as-deposited copper sulfide. The Gaussian width of peak $\text{Cu}_{\text{S(I)}}$ and $\text{Cu}_{\text{a(I)}}$ was systematically varied and the resultant reduced chi squared values for each fit are plotted against the respective width. The optimal peak-fitting results are presented in Fig. 2-A and Table 1.

The uniqueness plot corresponding to the peak-fitting procedure done on the Cu 2p photoemission spectra of the annealed copper sulfide film is presented in Fig. A2. The reduced chi squared values are determined from fittings obtained by varying, the same as before, the Gaussian width of peak $\text{Cu}_{\text{S(I)'}}$ and $\text{Cu}_{\text{a(I)'}}$. It can be observed that, although with a lower degree of curvature when compared to the plot from Fig. A1, the reduced chi squared value reaches a minimum at the optimal value of 0.42 eV. The lower degree of curvature is related to the presence of several overlapping peaks that compensate for each other as the Gaussian width is systematically varied, nevertheless, the peak-fitting appears to be appropriate.

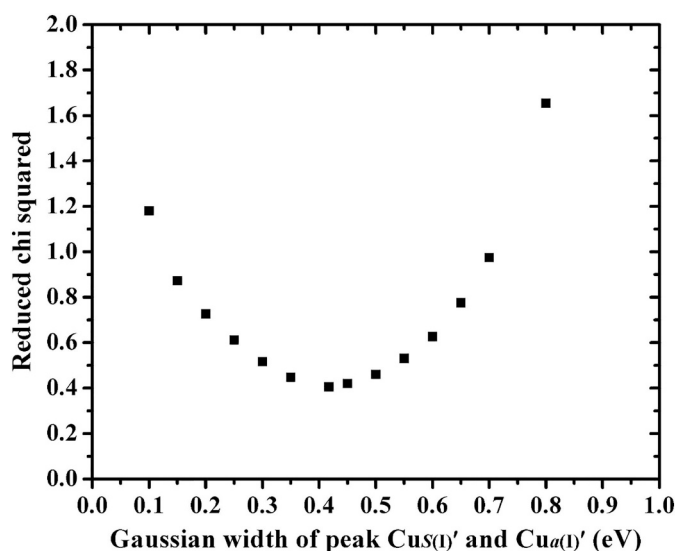


Fig. A2. Uniqueness plot obtained from peak-fitting the Cu 2p photoemission spectra of the annealed copper sulfide. The reduced chi squared values are plotted against the Gaussian width of peaks $\text{Cu}_{\text{S(I)'}}$ and $\text{Cu}_{\text{a(I)'}}$, which were systematically varied. The optimal peak-fitting results are presented in Fig. 4-A and Table 3.

The uniqueness plot obtained from peak-fittings of the S 2p photoemission spectrum of the as-deposited copper sulfide film is presented in Fig. A3. The parameter chosen to be systematically varied was the Gaussian width of the peaks S_1 and S_2 . The results show that the reduced chi squared values reach a minimal value at the optimal Gaussian width of 0.77 eV. Therefore, the approach employed to peak-fit this spectrum produces results that possess a good statistical quality.

The assessment of the peak-fitting, using a uniqueness plot, of the S 2p photoemission spectrum of the annealed copper sulfide film is presented in Fig. A4. The Gaussian width of the S_1' and S_2' peaks is fixed at different values within the 0.30–0.90 eV range. It is clear that the reduced chi squared value reaches a minimal value at 0.53 eV and the uniqueness plot shows a sharp curvature implying that the peak-fitting results presented Fig. 4-C and Table 3 have a low degree of fit parameter correlation.

Table 3

Peak parameters employed in the peak-fitting analysis of the photoemission spectra of the annealed copper sulfide film. The calculated uncertainties have a 95% confidence.

Core level	Peak label	$2p_{3/2}$ BE (eV)	Spin-orbit splitting (eV)	Peak width FWHM		SVSC (eV^{-1})	Normalized peak areas
				Gaussian (eV)	Lorentzian (eV)		
Cu 2p	Cu _{S(I)} '	931.93 ± 0.01	19.86 ± 0.02	0.42 ± 0.02	0.6	0.027	100 ± 3.9
	Cu _{a(I)} '	932.46 ± 0.05	21.05 ± 0.20	1.10 ± 0.05 ^a	0.6	0.027	17.5 ± 2.0
	Cu _{L3} '	949.67 ± 0.18	22.16 ± 0.20	0.42 ± 0.02	0.6	0.039	19.7 ± 2.0
	Cu _{L4} '	955.33 ± 0.24	19.97 ± 0.30	3.75 ± 0.50	–	0.039	18.0 ± 1.6
	Cu _{S(II)}	933.26 ± 0.09	19.61 ± 0.09	3.75 ± 0.50	–	0.041	72.5 ± 6.7
	Cu _{sat1(II)}	941.00 ± 0.19	20.27 ± 0.26	1.98 ± 0.17	0.6	0.041	21.8 ± 9.9
	Cu _{sat2(II)}	942.41 ± 0.09	19.94 ± 0.20	2.47 ± 0.50	0.6	0.041	4.7 ± 3.6
	Cu _{sat3(II)}	944.01 ± 0.04	18.87 ± 0.15	0.78 ± 0.30	0.6	0.041	19.2 ± 4.5
	Cu _{O1}	934.38 ± 0.17	20.19 ± 0.09	1.46 ± 0.13	0.6	0.041	62.4 ± 8.5
	Cu _{O2}	935.76 ± 0.17	21.98 ± 0.27	1.98 ± 0.17	0.6	0.041	11.4 ± 2.0
	Cu _{O3}	941.88 ± 0.50	20.43 ± 0.75	1.98 ± 0.17	0.6	0.041	18.7 ± 6.8
	Cu _{O4}	944.38 ± 0.49	18.06 ± 0.62	3.75 ± 0.50	0.6	0.041	12.5 ± 3.3
	S ₁ '	161.12 ± 0.02	1.20 ± 0.12	0.75 ± 0.35	0.6	0.012	65.5 ± 2.0
	S ₂ '	161.84 ± 0.02	1.26 ± 0.05	0.53 ± 0.05	0.07	0.012	100.0 ± 17.5
S 2p	S _d '	162.40 ± 0.45	1.40 ± 0.50	0.83 ± 0.30	0.07	0.012	42.5 ± 7.0
	S _{dd}	163.33 ± 0.21	1.10 ± 0.30	0.83 ± 0.30	0.07	0.012	37.7 ± 18.0
	S _x	165.25 ± 0.50	1.30 ± 0.70	1.44 ± 0.18	0.07	0.012	20.5 ± 5.5
	S _c	168.15 ± 0.16	1.20 ± 0.19	1.44 ± 0.18	0.07	0.012	33.0 ± 4.0

^a Value corresponding to the $2p_{1/2}$ photoemission line.

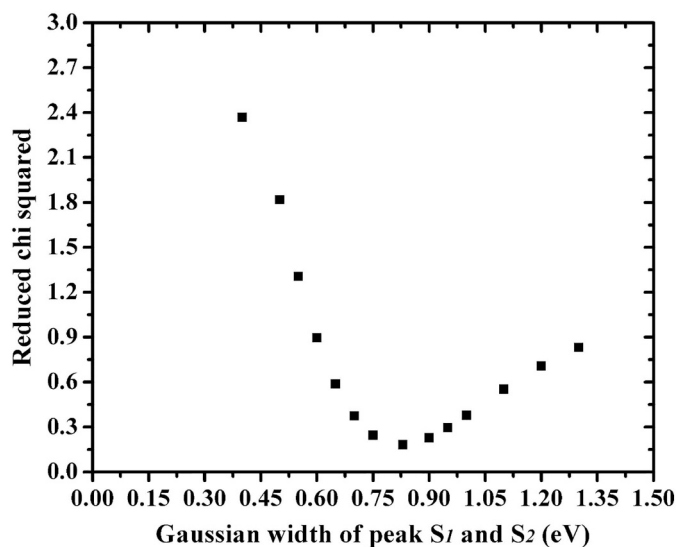


Fig. A3. Uniqueness plot obtained from peak-fitting the S 2p photoemission spectra of the as-deposited copper sulfide. Reduced chi squared values for several fittings are plotted against the Gaussian width of peak S₁ and S₂. The optimal peak-fitting results are presented in Fig. 2-C and Table 1.

Table 4

Physical parameters employed to calculate the chemical composition of copper sulfide films. The differential cross-section is determined using the relation reported by Herrera-Gomez [40] while the effective attenuation lengths are determined using the TPP-2M formula and the NIST Electron Inelastic Mean Free Path Database [42,43].

Core level	$\frac{d\sigma_{i,nl}}{d\Omega}$	Effective attenuation length ($\text{\AA}$)	
		CuS _x	CuO
Cu 2p	0.02707	10.3	9.3
S 2p	0.00179	21.3	19.4
O 1s	0.00314		14.6

Table 5

Atomic percent for each photoemission signal of the annealed film. The values were determined using Eq. (2). According to data presented below, the $(S_1' + S_2' + S_d' + S_{dd})/Cu_{S(1)'} + Cu_{a(1)'} + Cu_{S(II)} + Cu_{sat1(II)} + Cu_{sat2(II)} + Cu_{sat3(II)}$ atomic ratio, this is, the S/Cu ratio, is 1.00 ± 0.20 ; and the $(O_{Cu} + O_d) / (Cu_{O1} + Cu_{O2} + Cu_{O3} + Cu_{O4})$ atomic ratio, this is, the O/Cu ratio, is 1.29 ± 0.28 .

CuS _x layer ($d_s = \infty$)		Oxide layer ($d_{ox} = 4.5 \pm 0.8 \text{ \AA}$)	
Peak label	Atomic percent (%)	Peak label	Atomic percent (%)
$Cu_{S(1)'} + Cu_{a(1)'}$	24.93 ± 3.87	$Cu_{O1} + Cu_{O2} + Cu_{O3} + Cu_{O4}$	43.76 ± 10.00
$Cu_{S(II)} + Cu_{sat1(II)} + Cu_{sat2(II)} + Cu_{sat3(II)}$	25.07 ± 7.17	$O_{Cu} + O_d$	56.24 ± 8.00
S_1'	13.32 ± 2.05		
S_2'	20.34 ± 3.87		
S_d'	8.66 ± 1.63		
S_{dd}	7.68 ± 3.43		

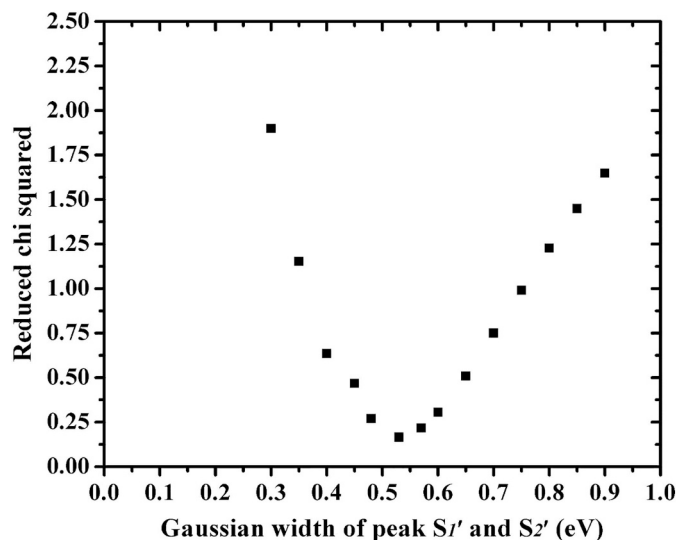


Fig. A4. Uniqueness plot obtained from peak-fitting the S 2p photoemission spectra of the annealed copper sulfide. Reduced chi squared values are obtained from fittings produced with distinct Gaussian widths for peaks S_1' and S_2' . The optimal peak-fitting results are presented in Fig. 4-C and Table 3.

Appendix B. Visual inspection of elements present at the surface

The survey scan spectrum of the as-deposited copper sulfide thin film is presented in Fig. B1. The peaks corresponding to Cu 3d, 3p, 3s, Auger, and 2p can be readily observed. Also, the corresponding S 2p and 2s peaks are clearly appreciated. Contamination related to the exposure of the films to air is corroborated by the presence of the C 1s and O 1s peaks, with part of these peaks being related to by-products of the deposition reaction. Regarding the latter, certain by-products of the deposition reaction are present on the surface of the films, that are related to sodium and nitrogen compounds due the presence of the Na 1s and N 1s peaks. It is possible that these by-products are related to Na_2SO_3 or sodium bonded with other oxoanion of sulfur, and the presence of nitrogen may be explained by the presence of cyanamide (H_2NCN) and the products of its subsequent reactions, such as cyanate, formed during the deposition of metal sulfide films when using thiourea as sulfur source [79].

The quantification of the sample must be done with caution, because by the sole use of peak area intensities, the amount of CuS_2 in the sample represents only 42.03%. Additionally, by visual inspection of the Cu 2p and S 2p signals, it can be appreciated that the background signal is larger when compared to the rest of the peaks related to oxygen, carbon, sodium, and nitrogen. These results, according to the visual inspection method reported by Tougaard [80], show that the elements identified in the survey spectrum are not homogeneously distributed in the film and as an initial rough estimation the elements O, C, Na, and N are surface localized with the CuS_2 film being located below the aforementioned elements. Therefore, the quantification of a possible surface layer on top of CuS_2 results in 42.84% of carbon, 23.03% of oxygen, 20.11% of nitrogen, 3.49% of sodium, and the inclusion of peaks S_x and S_c amount to 4.43 and 6.10%, respectively.

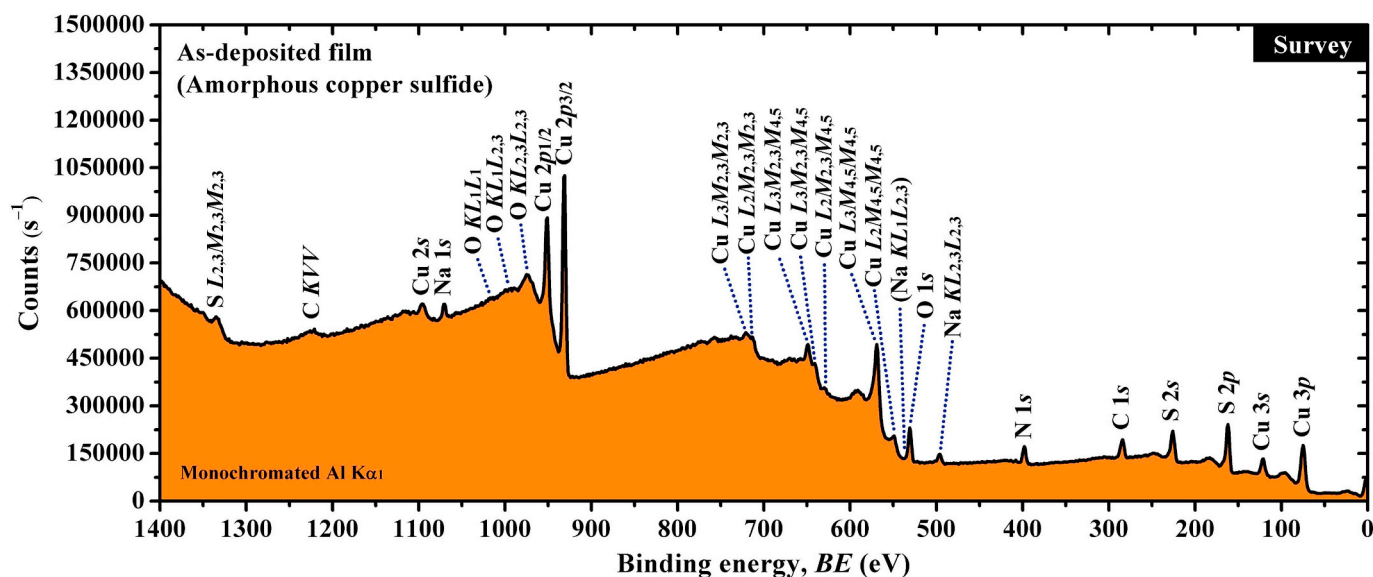


Fig. B1. X-ray photoelectron survey spectrum obtained from the as-deposited copper sulfide thin film. The core-level signals corresponding to copper, sulfur, oxygen, carbon, nitrogen, and sodium are clearly appreciated.

The survey scan spectrum of the annealed copper sulfide film is presented in Fig. B2. The same elements identified for the as-deposited film are present with varying intensity, meaning that during the annealing treatment there was no incorporation of additional elements to the films. As mentioned before, and in the main text, the CuS film is buried below an oxide layer and surface contamination that attenuates the photoemission signals related to this compound. Therefore, a simple peak area quantification is not reliable and representative of the sample, because it results that CuS represents 15.14% of the film. From visual inspection [80] of the Cu 2p and the S 2p photoemission peaks it can be implied that the large background signal is a clear indication that the annealed film has surface components. Therefore, the annealed film has a layered surface structure that can be represented by CuS located below, an oxide layer located in the middle (which are specified in the main text) and an outermost layer composed of 51.45% of carbon, 36.94% of oxygen, 3.51% of nitrogen, 2.46% of sodium, and the surface peaks S_x and S_c equal to 2.12 and 3.52%, respectively.

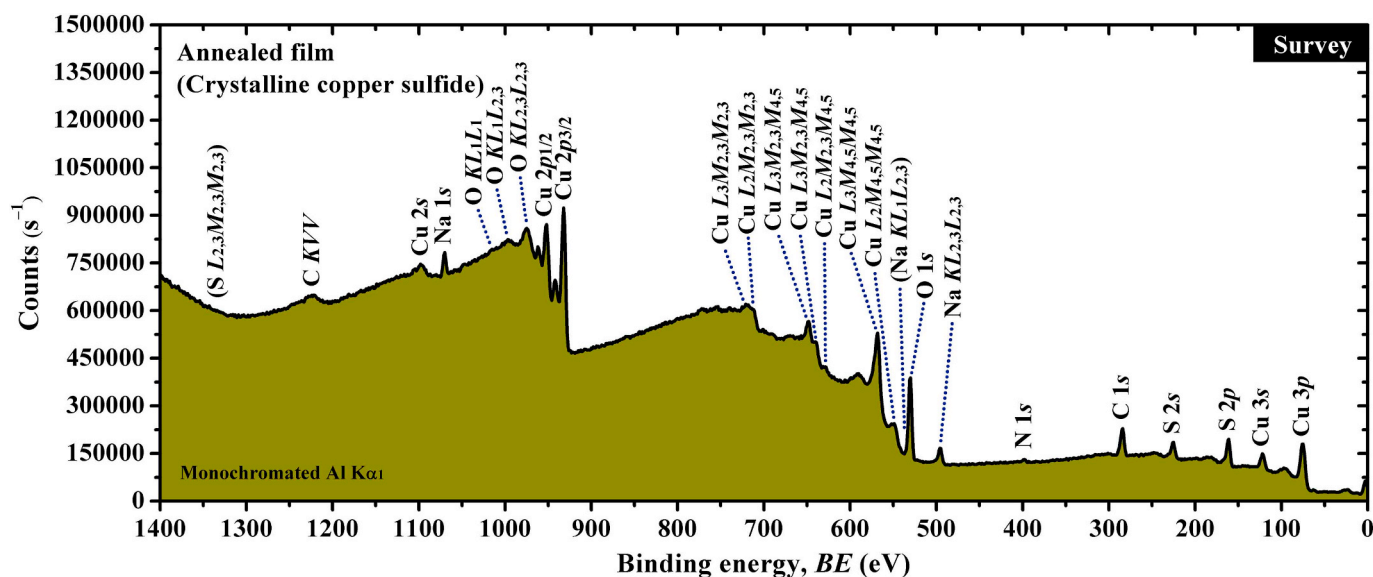


Fig. B2. X-ray photoelectron survey spectrum obtained from the annealed copper sulfide thin film. The peaks corresponding to copper, sulfur, oxygen, carbon, nitrogen, and sodium are clearly assigned to the respective photoemission signals.

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